CRYPTOGRAPHY AND NETWORK SECURITY

Assessment Tool 1

Course Code: CSA5112

Course Name: Cryptography and Network Security

Course Outcome Covered:

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for ensuring data security.

STUDENT DECLARATION

I, __E. Lakshmi Narayanan(192524249)__ (Name / Reg No), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student :

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Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

   a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
   b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

   a. Deduce the most likely original message.
   b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

   a. Demonstrate how the encryption process transforms the message step-by-step.
   b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

   a. Analyze how investigators could detect the presence of hidden data within the image.
   b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEEFEAOCAIVDEN”.

   a. Determine the most probable original message by reversing the encryption process.

b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

QUESTION 1: Event Communication Security

Objective

To understand the working of the Caesar Cipher using a fixed shift value.

Given Message

MEET AT NOON

a) Encryption using Shift Value = 4

Plain Text	**M E E T A T N O O N**
Cipher Text	Q I I X E X R S S R

Encrypted Message

QIIX EX RSSR

b) Decryption Process

The receiver knows that a shift of 4 positions was used.

Example:

- Q → M
- I → E
- X → T

By shifting every ciphertext letter 4 positions backward, the original message is obtained.

Original Message

MEET AT NOON

Result Analysis

The Caesar Cipher is easy to implement and suitable for basic message confidentiality. However, it provides limited security because attackers can easily perform brute-force attacks by trying all possible shifts.

Conclusion

The Caesar Cipher successfully encrypts and decrypts messages using a fixed key, demonstrating the fundamentals of classical cryptography.

QUESTION 2: Intercepted Suspicious Message

Objective

To analyze and decrypt an intercepted ciphertext.

Ciphertext

GSRH RH Z HVXIVG

a) Original Message

This ciphertext follows the **Atbash Cipher**, where letters are reversed in the alphabet.

Decrypting the entire message:

GSRH RH Z HVXIVG

becomes

THIS IS A SECRET

b) Justification

Observations:

1. The pattern contains repeated letters.
2. The word "RH" appears twice and maps correctly to "IS".
3. The Atbash Cipher uses reversed alphabet mapping:
   - A ↔ Z
   - B ↔ Y
   - C ↔ X
   - D ↔ W

Result Analysis

Pattern recognition and frequency analysis help identify simple substitution ciphers.

Conclusion

The intercepted message was successfully decrypted as:

THIS IS A SECRET

QUESTION 3: Secure Messaging System

Objective

To understand Vigenère Cipher encryption using a keyword.

Plaintext

HELLO

Keyword

KEY

Step 1: Repeat Keyword

Plaintext	**H**	**E**	**L**	**L**	**O**
Keyword	K	E	Y	K	E

Step 2: Convert to Numerical Values

Letter	**Value**
H	7
E	4
L	11
L	11
O	14

Keyword Values:

Letter	**Value**
K	10
E	4
Y	24
K	10
E	4

Step 3: Encryption

Formula:

Cipher = (Plaintext + Key) mod 26

Plain	**Key**	**Result**	**Cipher**
7	10	17	R
4	4	8	I
11	24	9	J

Plain	Key	Result	Cipher
11	10	21	V
14	4	18	S

Ciphertext

RIJVS

b) Influence of Repeating Keyword

The keyword repeats continuously and determines the shift value applied to each character.

KEYKEYKEY...

Each plaintext letter is encrypted differently depending on the corresponding keyword character, making the cipher stronger than Caesar Cipher.

Result Analysis

The Vigenère Cipher reduces vulnerability to frequency analysis because multiple shifts are used.

Conclusion

Keyword repetition enhances confidentiality by creating varying encryption patterns.

QUESTION 4: Suspicious Digital Asset Investigation

Objective

To analyze hidden information inside digital images using steganography.

a) Detecting Hidden Data

Investigators can use:

1. Statistical Analysis

- Examine unusual pixel patterns.
- Detect changes in image characteristics.

2. LSB (Least Significant Bit) Analysis

- Analyze least significant bits of pixels.
- Hidden information is often stored here.

3. Steganalysis Tools

Examples:

- StegExpose
- StegSecret
- OpenStego

These tools help identify hidden content.

b) Techniques to Improve Security

Technique 1: Encrypt Before Hiding

- Apply AES encryption.
- Then embed encrypted data into the image.

Benefit:
Even if extracted, data remains unreadable.

Technique 2: Randomized Embedding

- Hide information in random pixel locations.
- Use a secret key for positioning.

Benefit:

Makes detection significantly harder.

Result Analysis

Combining cryptography and steganography provides two layers of security.

Conclusion

Steganographic communications become more secure when encryption and randomized embedding are used.

QUESTION 5: Legacy Encryption in Corporate Communication

Objective

To analyze Rail Fence Transposition Cipher and evaluate its security.

Ciphertext

WECRLTEERDSOEEFEAOCAIVDEN

a) Original Message

This ciphertext is produced using a **3-Rail Rail Fence Cipher**.

Reconstructing the rails:

Rail 1: W . . . E . . . C . . . R . . . L . . . T . . . E

Rail 2: . E . R . D . S . O . E . E . F . E . A . O . C

Rail 3: . . A . . . I . . . V . . . D . . . E . . . N

Reading in zig-zag order gives:

Original Message

WE ARE DISCOVERED FLEE AT ONCE

b) Vulnerabilities

1. No substitution of characters.
2. Character frequencies remain unchanged.
3. Easily broken using pattern analysis.
4. Susceptible to brute-force attacks.
5. Provides weak protection against modern computing power.

c) Practical Enhancement

Implement **AES (Advanced Encryption Standard)** encryption before applying transposition.

Benefits:

- Strong confidentiality
- Resistant to brute-force attacks
- Widely accepted industry standard

Result Analysis

Modern encryption algorithms provide significantly stronger protection than classical transposition methods.

Conclusion

The Rail Fence Cipher demonstrates basic transposition principles but is inadequate for modern secure communication systems.